

Digital Logic and Design

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Lecture No. 09

Recap

- Karnaugh Maps
- Mapping Standard SOP expressions
- Mapping Non-Standard SOP expressions
- Simplification of K-maps
- Don't care states

Mapping a Standard POS expression

- Selecting n-variable K-map
- 0 marked in cell for each maxterm
- Remaining cells marked with 1

Mapping of Standard POS expression

- POS expression

$$(A + B + \overline{C}).(A + \overline{B} + C).(\overline{A} + B + \overline{C}).(\overline{A} + \overline{B} + \overline{C})$$

Mapping of Standard POS expression

- POS expression

$$(A + B + \bar{C}).(A + \bar{B} + C).(\bar{A} + B + \bar{C}).(\bar{A} + \bar{B} + \bar{C})$$

AB\C	0	1
00	1	0
01	0	1
11	1	0
10	1	0

A\BC	00	01	11	10
0	1	0	1	0
1	1	0	0	1

Simplification of POS expressions using K-map

- Mapping of expression
- Forming of Groups of 0s
- Each group represents sum term
- 3-variable K-map
 - 1 cell group yields a 3 variable sum term
 - 2 cell group yields a 2 variable sum term
 - 4 cell group yields a 1 variable sum term
 - 8 cell group yields a value of 0 for function

Simplification of POS expressions using K-map3

- 4-variable K-map
 - 1 cell group yields a 4 variable sum term
 - 2 cell group yields a 3 variable sum term
 - 4 cell group yields a 2 variable sum term
 - 8 cell group yields a 1 variable sum term
 - 16 cell group yields a value of 0 for function

Simplification of POS expressions using K-map

$$(B + C).(A + \bar{B} + \bar{C})$$

AB\C	0	1
00	0	1
01	1	0
11	1	1
10	0	1

A\BC	00	01	11	10
0	0	1	1	1
1	1	0	0	0

$$(A + B + C).(\bar{A} + \bar{C}).(\bar{A} + \bar{B})$$

Simplification of POS expressions using K-map

$$(B + C).(A + \bar{B} + \bar{C})$$

AB\C	0	1
00	0	1
01	1	0
11	1	1
10	0	1

A\BC	00	01	11	10
0	0	1	1	1
1	1	0	0	0

$$(A + B + C).(\bar{A} + \bar{C}).(\bar{A} + \bar{B})$$

Simplification of POS expressions using K-map

$$(A + B).(B + C)$$

AB\C	0	1
00	0	0
01	1	1
11	1	1
10	0	1

A\BC	00	01	11	10
0	0	0	1	1
1	1	1	1	0

$$(A + B).(\bar{A} + \bar{B} + C)$$

Simplification of POS expressions using K-map

$$(A + \bar{B} + C).(A + C + D).(B + \bar{C} + D)$$

AB\CD	00	01	11	10
00	0	1	1	0
01	0	0	1	1
11	1	1	1	1
10	1	1	1	0

Simplification of POS expressions using K-map

$$(A + C).(C + \bar{D}).(B + \bar{C} + D)$$

AB\CD	00	01	11	10
00	0	0	1	0
01	0	0	1	1
11	1	0	1	1
10	1	0	1	0

Simplification of POS expressions using K-map

$$(A + \bar{B} + C).(A + \bar{B} + \bar{D}).(B + C + \bar{D}).(\bar{A} + \bar{B} + \bar{C} + D)$$

AB\CD	00	01	11	10
00	1	0	1	1
01	0	0	0	1
11	1	1	1	0
10	1	0	1	1

Conversion between SOP & POS using K-map

- Groups of 1s represents SOP expression
- Groups of 0s represents POS expression

Conversion between SOP & POS using K-map

$$\overline{B}\overline{D} + \overline{B}C + A\overline{B}\overline{C} + ABD + \overline{A}C\overline{D}$$

AB\CD	00	01	11	10
00	1	0	1	1
01	0	0	0	1
11	1	1	1	0
10	1	0	1	1

5-Variable K-map

- Represented as two, 4 variable K-map

5-Variable K-map

$A = 0$

BC\DE	00	01	11	10
00	0	1	3	2
01	4	5	7	6
11	12	13	15	14
10	8	9	11	10

5-Variable K-map

$A = 1$

BC\DE	00	01	11	10
00	16	17	19	18
01	20	21	23	22
11	28	29	31	30
10	24	25	27	26

Simplification of a 5-Variable K-map

- 5 variable K-map mapping
- 5 variable K-map grouping
- 5 variable K-map simplification

5-Variable K-map simplification

$$A = 0$$

BC\DE	00	01	11	10
00	0	1	0	1
01	0	1	0	0
11	0	0	0	1
10	0	0	1	1

5-Variable K-map simplification

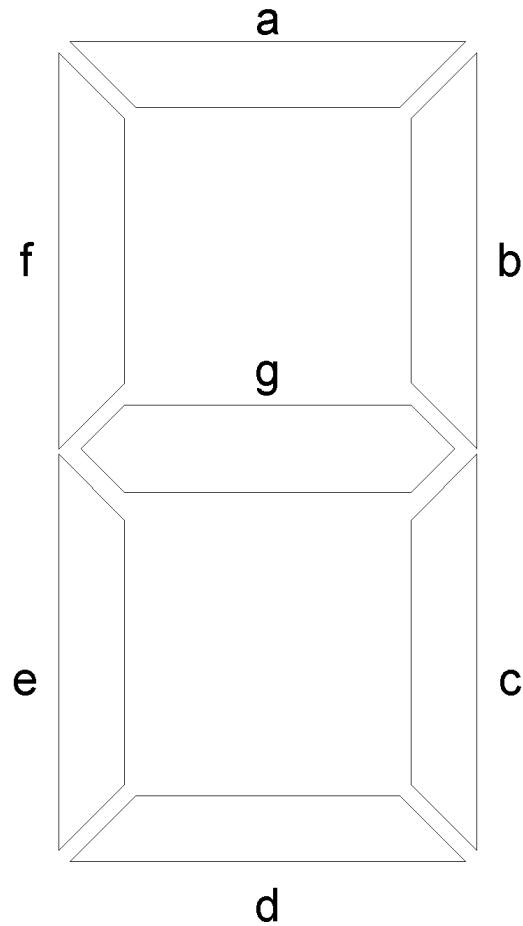
$A = 1$

BC\DE	00	01	11	10
00	1	1	0	0
01	1	1	0	0
11	0	0	0	1
10	0	1	1	1

Functions having multiple outputs

- Ckt receives a BCD number input
- Displays decimal number 0 to 9 on a single digit 7-segment display
- Ckt receives two 2-bit numbers A and B
- Sets one of three outputs to $>$, $=$, or $<$

7-Segment Display



Function Table for Segment 'a'

Inputs				Output
A	B	C	D	a
0	0	0	0	1
0	0	0	1	0
0	0	1	0	1
0	0	1	1	1
0	1	0	0	0
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1

Inputs				Output
A	B	C	D	a
1	0	0	0	1
1	0	0	1	1
1	0	1	0	X
1	0	1	1	X
1	1	0	0	X
1	1	0	1	X
1	1	1	0	X
1	1	1	1	X

Function Table for Segment 'b'

Inputs				Output
A	B	C	D	b
0	0	0	0	1
0	0	0	1	1
0	0	1	0	1
0	0	1	1	1
0	1	0	0	1
0	1	0	1	0
0	1	1	0	0
0	1	1	1	1

Inputs				Output
A	B	C	D	b
1	0	0	0	1
1	0	0	1	1
1	0	1	0	X
1	0	1	1	X
1	1	0	0	X
1	1	0	1	X
1	1	1	0	X
1	1	1	1	X

Function Table for Segment 'c'

Inputs				Output
A	B	C	D	c
0	0	0	0	1
0	0	0	1	1
0	0	1	0	0
0	0	1	1	1
0	1	0	0	1
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1

Inputs				Output
A	B	C	D	c
1	0	0	0	1
1	0	0	1	1
1	0	1	0	X
1	0	1	1	X
1	1	0	0	X
1	1	0	1	X
1	1	1	0	X
1	1	1	1	X

Function Table for Segment 'd'

Inputs				Output
A	B	C	D	d
0	0	0	0	1
0	0	0	1	0
0	0	1	0	1
0	0	1	1	1
0	1	0	0	0
0	1	0	1	1
0	1	1	0	1
0	1	1	1	0

Inputs				Output
A	B	C	D	d
1	0	0	0	1
1	0	0	1	1
1	0	1	0	X
1	0	1	1	X
1	1	0	0	X
1	1	0	1	X
1	1	1	0	X
1	1	1	1	X

Function Table for Segment 'e'

Inputs				Output
A	B	C	D	e
0	0	0	0	1
0	0	0	1	0
0	0	1	0	1
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	1
0	1	1	1	0

Inputs				Output
A	B	C	D	e
1	0	0	0	1
1	0	0	1	0
1	0	1	0	X
1	0	1	1	X
1	1	0	0	X
1	1	0	1	X
1	1	1	0	X
1	1	1	1	X

Function Table for Segment 'f'

Inputs				Output
A	B	C	D	f
0	0	0	0	1
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	1
0	1	0	1	1
0	1	1	0	1
0	1	1	1	0

Inputs				Output
A	B	C	D	f
1	0	0	0	1
1	0	0	1	1
1	0	1	0	X
1	0	1	1	X
1	1	0	0	X
1	1	0	1	X
1	1	1	0	X
1	1	1	1	X

Function Table for Segment 'g'

Inputs				Output
A	B	C	D	g
0	0	0	0	0
0	0	0	1	0
0	0	1	0	1
0	0	1	1	1
0	1	0	0	1
0	1	0	1	1
0	1	1	0	1
0	1	1	1	0

Inputs				Output
A	B	C	D	g
1	0	0	0	1
1	0	0	1	1
1	0	1	0	X
1	0	1	1	X
1	1	0	0	X
1	1	0	1	X
1	1	1	0	X
1	1	1	1	X

Karnaugh Map for Segment 'a'

$$A + C + BD + \overline{B}\overline{D}$$

AB\CD	00	01	11	10
00	1	0	1	1
01	0	1	1	1
11	x	x	x	x
10	1	1	x	x

The Karnaugh map for Segment 'a' is a 5x5 grid. The columns are labeled AB\CD (00, 01, 11, 10) and the rows are labeled 00, 01, 11, 10. The cells contain the following values: (00,00)=1, (00,01)=0, (00,11)=1, (00,10)=1, (01,00)=0, (01,01)=1, (01,11)=1, (01,10)=1, (11,00)=x, (11,01)=x, (11,11)=x, (11,10)=x, (10,00)=1, (10,01)=1, (10,11)=x, (10,10)=x. There are four groupings: a group of two cells (00,00) and (00,10) circled; a group of four cells (00,11), (01,11), (10,11), and (11,11) circled; a group of four cells (01,01), (01,11), (10,01), and (10,11) circled; and a group of four cells (00,10), (01,10), (10,00), and (10,10) circled.

Karnaugh Map for Segment 'b'

$$\overline{B} + \overline{C}\overline{D} + CD$$

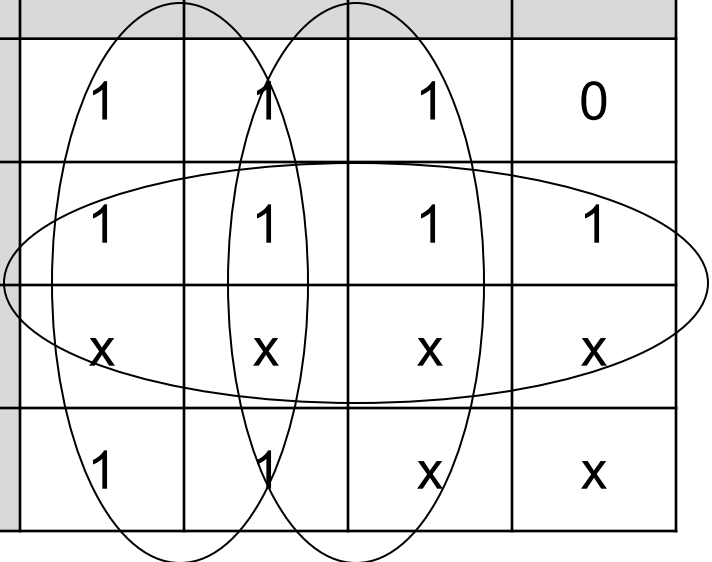
AB\CD	00	01	11	10
00	1	1	1	1
01	1	0	1	0
11	x	x	x	x
10	1	1	x	x

The Karnaugh map for Segment 'b' is a 4x4 grid. The columns are labeled 00, 01, 11, and 10. The rows are labeled 00, 01, 11, and 10. The cells contain the following values: (00,00)=1, (01,00)=1, (11,00)=1, (10,00)=1; (00,01)=1, (01,01)=0, (11,01)=1, (10,01)=0; (00,11)=x, (01,11)=x, (11,11)=x, (10,11)=x; (00,10)=1, (01,10)=1, (11,10)=x, (10,10)=x. There are three groupings indicated by hand-drawn lines: a group of four 1s in the top row (00,00), (01,00), (11,00), (10,00); a group of four 1s in the first column (00,00), (00,01), (00,10), (00,11); and a group of four 1s in the third column (11,00), (11,01), (11,10), (11,11).

Karnaugh Map for Segment 'c'

$$\overline{C} + D + B$$

AB\CD	00	01	11	10
00	1	1	1	0
01	1	1	1	1
11	x	x	x	x
10	1	1	x	x



The Karnaugh map for segment 'c' is a 5x5 grid. The first column (AB\CD) and the first two rows (00, 01) are shaded gray. The map contains 1s in cells (00,00), (01,00), (11,00), (00,01), (01,01), (11,01), (00,10), (01,10), and (10,10). It contains x's in cells (11,00), (10,00), (11,01), (10,01), (11,10), (10,10), (11,11), (10,11), and (11,12). Three groupings are shown: a horizontal group of four 1s in the first row (00,00) to (11,00); a vertical group of four 1s in the first column (00,00) to (10,10); and a group of four 1s in the second row (01,01) to (10,10).

Karnaugh Map for Segment 'd'

$$A + \overline{B}\overline{D} + \overline{B}C + C\overline{D} + B\overline{C}D$$

AB\CD	00	01	11	10
00	1	0	1	1
01	0	1	0	1
11	x	x	x	x
10	1	1	x	x

Karnaugh Map for Segment 'e'

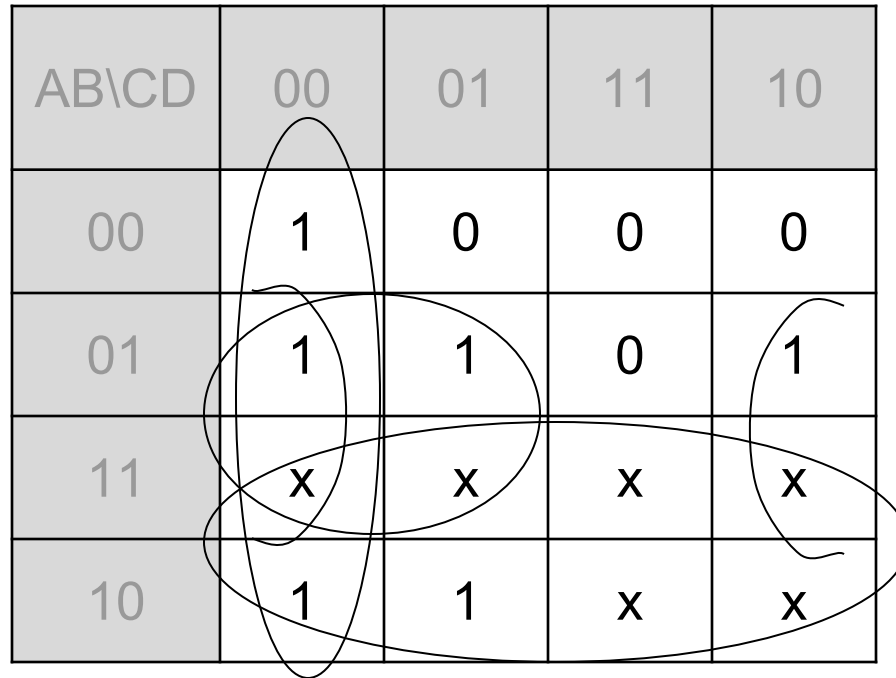
$$\overline{B}\overline{D} + C\overline{D}$$

AB\CD	00	01	11	10
00	1	0	0	1
01	0	0	0	1
11	x	x	x	x
10	1	0	x	x

Karnaugh Map for Segment 'f'

$$B + \overline{C}\overline{D} + B\overline{C} + B\overline{D}$$

AB\CD	00	01	11	10
00	1	0	0	0
01	1	1	0	1
11	x	x	x	x
10	1	1	x	x



The Karnaugh map for segment 'f' is a 4x4 grid. The columns are labeled 00, 01, 11, and 10. The rows are labeled 00, 01, 11, and 10. The cells contain the following values: (00,00)=1, (00,01)=0, (00,11)=0, (00,10)=0; (01,00)=1, (01,01)=1, (01,11)=0, (01,10)=1; (11,00)=x, (11,01)=x, (11,11)=x, (11,10)=x; (10,00)=1, (10,01)=1, (10,11)=x, (10,10)=x. There are four groupings indicated by hand-drawn lines: a vertical group of four cells (00,00), (01,00), (00,01), (01,01) for the term B; a horizontal group of four cells (00,01), (01,01), (11,01), (10,01) for the term C̄D̄; a horizontal group of four cells (00,01), (01,01), (00,11), (01,11) for the term B̄C̄; and a horizontal group of four cells (00,11), (01,11), (11,11), (10,11) for the term B̄D̄.

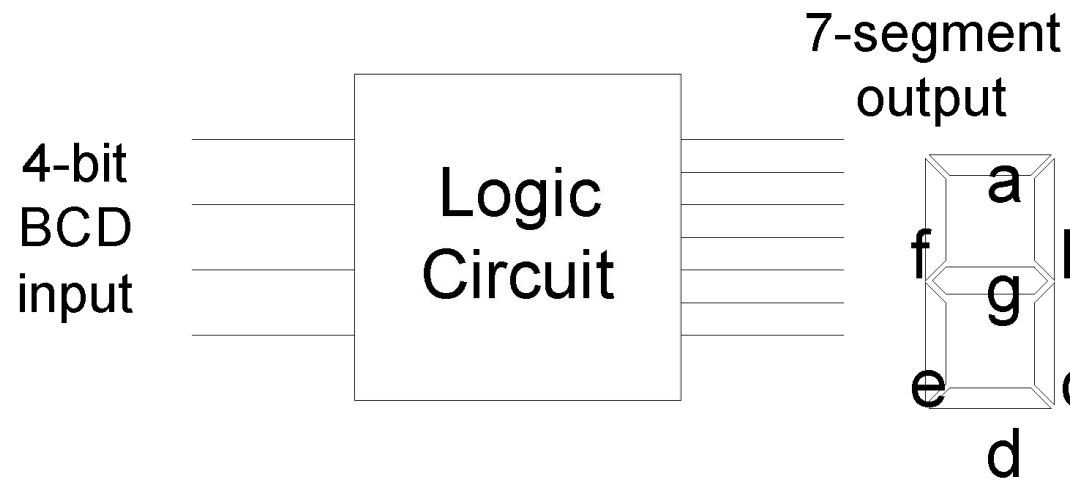
Karnaugh Map for Segment 'g'

$$A + B\bar{C} + C\bar{D} + \bar{B}C$$

AB\CD	00	01	11	10
00	0	0	1	1
01	1	1	0	1
11	x	x	x	x
10	1	1	x	x

The Karnaugh map for segment 'g' is a 5x5 grid. The rows are labeled AB\CD (00, 01, 11, 10) and the columns are labeled 00, 01, 11, 10. The cells contain values: 0, 0, 1, 1 for row 00; 1, 1, 0, 1 for row 01; x, x, x, x for row 11; and 1, 1, x, x for row 10. Hand-drawn groupings include: a horizontal group of 1s in row 00 (columns 11 and 10); a horizontal group of 1s in row 01 (columns 00 and 01); a horizontal group of 1s in row 10 (columns 00 and 01); a vertical group of 1s in column 01 (rows 00 and 01); a vertical group of 1s in column 10 (rows 00 and 01); a horizontal group of x's in row 11 (columns 00 and 01); a horizontal group of x's in row 10 (columns 00 and 01); a vertical group of x's in column 11 (rows 00 and 01); a vertical group of x's in column 10 (rows 00 and 01); a large group of 1s and x's in column 00 (rows 00, 01, 10); a large group of 1s and x's in column 01 (rows 00, 01, 10); and a large group of 1s and x's in column 10 (rows 00, 01, 10).

7-Segment Circuit



Comparator Circuit

- Inputs two 2-bit binary numbers A and B
- Has three outputs
- $A > B$
- $A = B$
- $A < B$

Function Table for $A > B$

Inputs				Output
A_1	A_0	B_1	B_0	$A > B$
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	1
0	1	0	1	0
0	1	1	0	0
0	1	1	1	0

Inputs				Output
A_1	A_0	B_1	B_0	$A > B$
1	0	0	0	1
1	0	0	1	1
1	0	1	0	0
1	0	1	1	0
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	0

Function Table for $A=B$

Inputs				Output
A_1	A_0	B_1	B_0	$A=B$
0	0	0	0	1
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	1
0	1	1	0	0
0	1	1	1	0

Inputs				Output
A_1	A_0	B_1	B_0	$A=B$
1	0	0	0	0
1	0	0	1	0
1	0	1	0	1
1	0	1	1	0
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	1

Function Table for $A < B$

Inputs				Output
A_1	A_0	B_1	B_0	$A < B$
0	0	0	0	0
0	0	0	1	1
0	0	1	0	1
0	0	1	1	1
0	1	0	0	0
0	1	0	1	0
0	1	1	0	1
0	1	1	1	1

Inputs				Output
A_1	A_0	B_1	B_0	$A < B$
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	1
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	0

Karnaugh Map for $A > B$

$$A_1 \overline{B_1} + A_0 \overline{B_1} \overline{B_0} + A_1 A_0 \overline{B_0}$$

$A_1 A_0 / B_1 B_0$	00	01	11	10
00	0	0	0	0
01	1	0	0	0
11	1	1	0	1
10	1	1	0	0

Karnaugh Map for A=B

$$\overline{A_1}\overline{A_0}\overline{B_1}\overline{B_0} + \overline{A_1}A_0\overline{B_1}B_0 + A_1A_0B_1B_0 + A_1\overline{A_0}B_1\overline{B_0}$$

A_1A_0 / B_1B_0	00	01	11	10
00	1	0	0	0
01	0	1	0	0
11	0	0	1	0
10	0	0	0	1

Karnaugh Map for $A < B$

$$\overline{A_1}B_1 + \overline{A_1}\overline{A_0}B_0 + \overline{A_0}B_1B_0$$

A_1A_0 / B_1B_0	00	01	11	10
00	0	1	1	1
01	0	0	1	1
11	0	0	0	0
10	0	0	1	0

Odd Prime Number Circuit

- Circuit detects odd prime numbers for a 5-bit input number
- Function represented by minterms
3, 5, 7, 11, 13, 17, 19, 23, 29 and 31